

# ReadMate

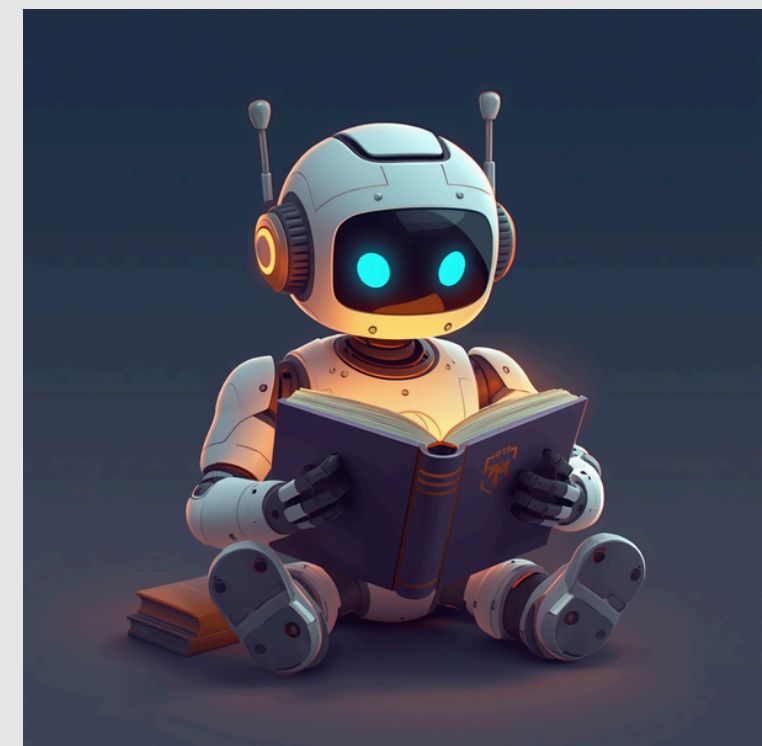
**Big data analytics and recommendation System for books**

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Module name :IT5612 - Big Data Analytics for Artificial Intelligence

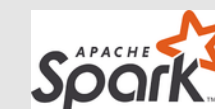
# Overview

This project not only uses large scale Goodreads data to find useful insights, but also a real recommendation technique with ALS collaborative filtering to make personalized book recommendations. The workflow, outputs of analytics, model training, evaluation, and recommendations are clearly explained and demonstrated using the Streamlit system interface, and scalable big data processing is performed using Apache Spark and PySpark.



## Objectives

- Analyze large-scale Goodreads datasets
- Perform big data analytics
- Build personalized recommendation system
- Generate recommendations for:
  - existing users
  - new users





# Dataset Description

Link to the DataSet : <https://cseweb.ucsd.edu/~jmcauley/datasets/goodreads.html>



## Dataset Size

- ~258K books
- ~3.4M reviews

Research data from the Goodreads Book Graph Dataset include millions of books, interactions between users and these books, and Goodreads reviews for academic purposes. Includes metadata for the books, ratings, reviews, and genre subsets for recommendations and NLP experiments.

```
Reviews Dataset Schema:
root
|-- book_id: string (nullable = true)
|-- date_added: string (nullable = true)
|-- date_updated: string (nullable = true)
|-- n_comments: long (nullable = true)
|-- n_votes: long (nullable = true)
|-- rating: long (nullable = true)
|-- read_at: string (nullable = true)
|-- review_id: string (nullable = true)
|-- review_text: string (nullable = true)
|-- started_at: string (nullable = true)
|-- user_id: string (nullable = true)

Books Dataset Schema:
root
|-- asin: string (nullable = true)
|-- authors: array (nullable = true)
|   |-- element: struct (containsNull = true)
|   |   |-- author_id: string (nullable = true)
|   |   |-- role: string (nullable = true)
|-- average_rating: string (nullable = true)
|-- book_id: string (nullable = true)
|-- country_code: string (nullable = true)
|-- description: string (nullable = true)
```

The project uses two Goodreads datasets.

- The first dataset is `goodreads_books_fantasy_paranormal.json.gz`, which contains book metadata such as title, average rating, publication year, ratings count, and review count.
- The second dataset is `goodreads_reviews_fantasy_paranormal.json.gz`, which contains user reviews and rating information.

# Workflow

This is a complete pipeline workflow of the Big Data analytics and recommendation system, which follows the requirement of the mini-project and shows how you can implement every stage of the pipeline using Apache Spark and PySpark.

- Loads and preprocesses large Goodreads dataset; Conducts exploratory analytics and visualizations;
- Performs feature engineering;
- Trains an ALS collaborative filtering recommendation model;
- Evaluates the model with RMSE and presents them in an interactive Streamlit interface to provide personalized book recommendations.





# Data Preprocessing & Feature Engineering

## Preprocessing

- Remove null values
- Remove duplicates
- Convert datatypes
- Join review and metadata datasets

## Feature Engineering

- user\_id → user\_index
- book\_id → book\_index
- rating matrix generation



## Feature Engineering

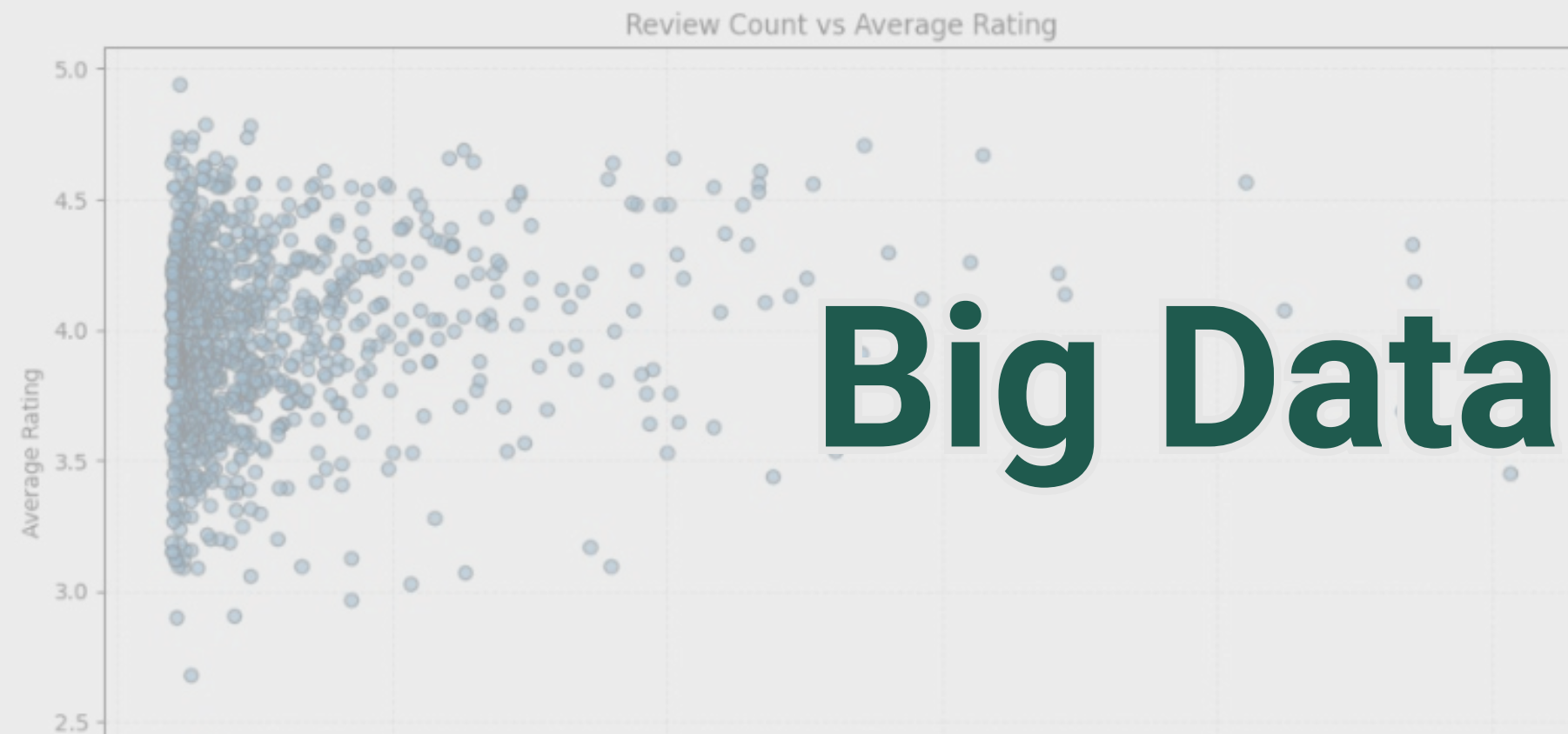
Feature engineering converts raw Goodreads data into a format suitable for machine learning. Spark MLlib ALS requires numeric user indexes, numeric item indexes, and rating values.

|   | Feature          | Purpose                                            |
|---|------------------|----------------------------------------------------|
| 0 | user_index       | Numeric user identifier used by ALS.               |
| 1 | book_index       | Numeric book identifier used by ALS.               |
| 2 | rating           | User rating used as model input.                   |
| 3 | book_id          | Original Goodreads book identifier.                |
| 4 | title            | Book title used for displaying recommendations.    |
| 5 | average_rating   | Book-level average rating used for interpretation. |
| 6 | ratings_count    | Popularity indicator based on number of ratings.   |
| 7 | publication_year | Used for trend analysis and display.               |
| 8 | item_factors     | Latent book feature vectors learned by ALS.        |

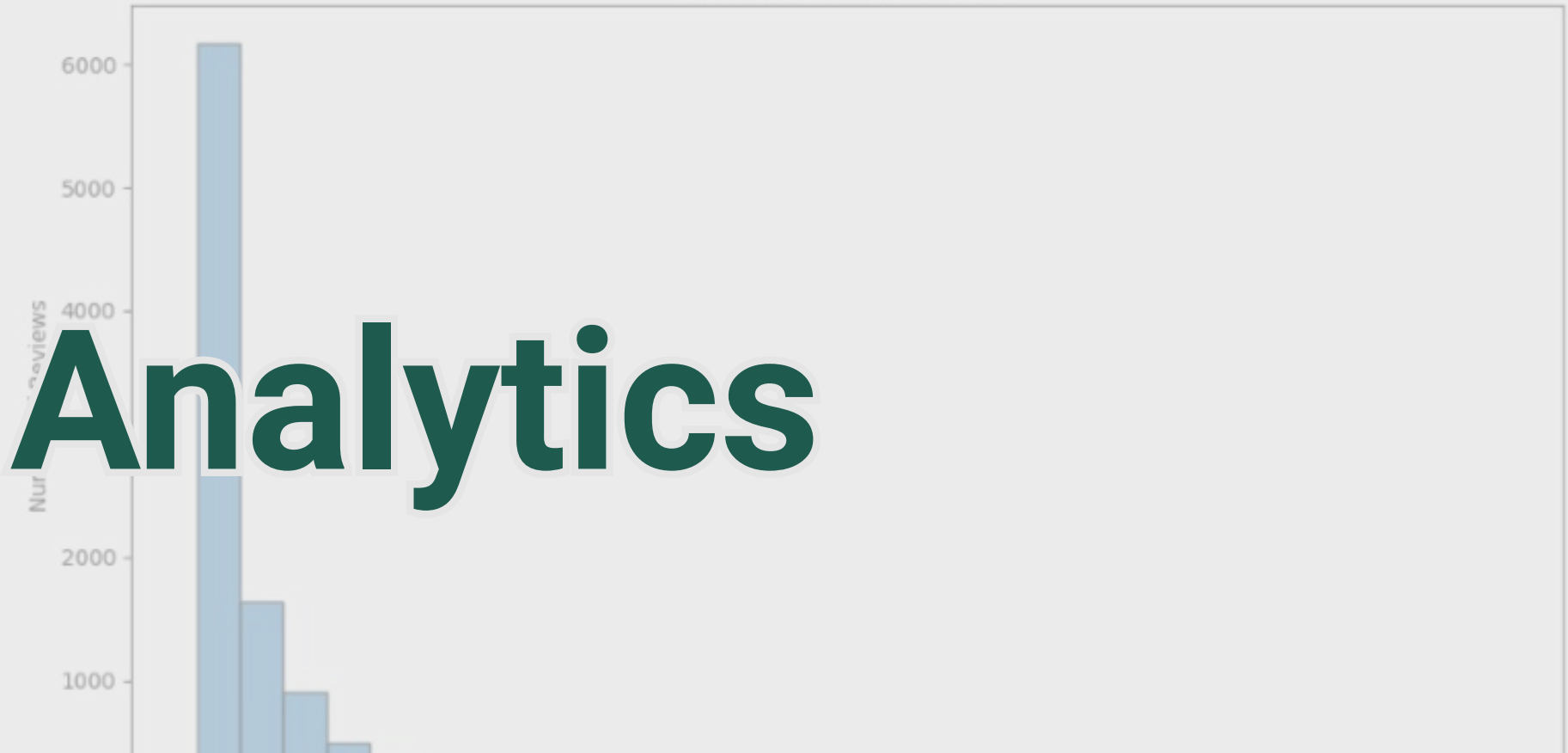
## ALS Input Format

```
user_id → user_index
book_id → book_index
rating → ALS input rating
```

Review Count vs Average Rating



Review Length Distribution



# Big Data Analytics

1

## Descriptive Analytics

Books By Publication Year

Rating Distribution

Top Reviewed Books

Review Count By Publication Year

Review Length Distribution

Purpose

Describes patterns in the data, user preferences, and popularity of books

2

## Diagnostic Analytics

Review Count Vs Average Rating

Purpose

Interpretes the connection between popularity and ratings by users to gain insight into engagement.

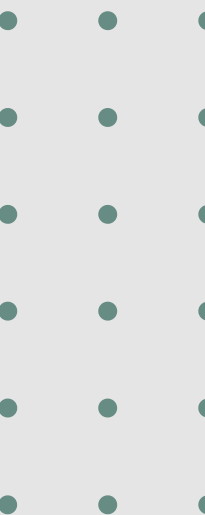
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## Predictive Analytics

Predicted Most Rateable Books

Purpose

Predicts future books that are likely to be very engaging and rated highly by users.







# The Recommendations

## 1. Existing User Recommendation

### ALS Collaborative Filtering

- User selects an existing user index
- ALS model analyzes previous rating behavior
- Generates personalized Top-N book recommendations
- Different users receive different recommendations based on interaction patterns

### Output

- Recommended books
- Predicted ratings
- Personalized suggestions

## 2. New User Recommendation

### Cold-Start Recommendation

- Handles users not available in the dataset
- New users select and rate books based on preferences
- System identifies highly rated and similar books
- Generates recommendations without historical user data

### Output

- Preference-based recommendations
- Similar and highly rated books
- Personalized suggestions for new users

 **ReadMate: Personalized Book Recommendation System**

Big Data Analytics and Recommendation System using Apache Spark

### Recommendation Results

This section displays personalized Top-N recommendations generated by the trained ALS model. Select a user index to view the books recommended for that user.

Select User Index

12565

|   | title                                                 | predicted_rating | average_rating | ratings_count | publication_year |  |
|---|-------------------------------------------------------|------------------|----------------|---------------|------------------|--|
| 0 | The Shadow of What Was Lost (The Licanus Trilogy, #1) | 6.5705           | 4.36           | 1005          | None             |  |
| 1 | ᠰᠠᠶᠢᠨ ᠤᠯᠤᠰ ᠤᠨ ᠤᠯᠤᠰ ᠤᠨ ᠤᠯᠤᠰ                            | 6.3865           | 4.45           | 7             | 2016             |  |
| 2 | Mine                                                  | 6.2804           | 4.17           | 94            | 2012             |  |
| 3 | Los sin rostro (Detective esquelito, #3)              | 6.1378           | 4.4            | 32            | 2009             |  |
| 4 | Graveyard (Ed & Lorraine Warren #1)                   | 6.1452           | 3.67           | 435           | None             |  |
| 5 | Archive of Fire (Sentinel #1)                         | 6.1353           | 4.22           | 5             | None             |  |
| 6 | Prince Elcheon and the Secret of the Ancient          | 6.1197           | 3.81           | 30            | 2008             |  |
| 7 | Fierce Heartbreaker (Sierra Pride, #2)                | 6.0966           | 4.15           | 33            | 2016             |  |
| 8 | Silo (Wool, #1)                                       | 6.0333           | 4.24           | 139           | 2014             |  |
| 9 | The Centaur's Bride: A Mates for Monsters Novella     | 5.9977           | 3.95           | 119           | None             |  |

 **ReadMate: Personalized Book Recommendation System**

Big Data Analytics and Recommendation System using Apache Spark

### New User Recommendations

This section allows a new user to receive personalized recommendations. The user selects a few books they have read and provides ratings. The system then builds a temporary user preference vector using trained ALS item factors and recommends books based on that vector.

Select up to 5 books you've read and rate them. The system will infer your preferences from the trained ALS item factors.

Select books for your profile

Harry Potter and... x The Hobbit & Th... x Harri

Select 11631 matches

Harrow

Harrowgate

Harrison Squared

Harrowing the Dragon

Harry's First Christmas

Harry Potter Film Wizardry

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Select up to 5 books you've read and rate them. The system will infer your preferences from the trained ALS item factors.

Select books for your profile

Harry Potter and... x The Hobbit & Th... x Harry Potter and... x George R.R. Mart... x Harry Potter and...

Rating for: Harry Potter and the Cursed Child

4.50

Rating for: The Hobbit & The Lord of the Rings

Rating for: Harry Potter and the Prisoner of Azkaban

4.50

Rating for: George R.R. Martin's Game Of Thrones #1

Rating for: Harry Potter and the Order of the Phoenix

3.50

Generate Top 5 Recommendations

|       | book_id  | title                                  | predicted_score | average_rating | ratings_count | publication_year |
|-------|----------|----------------------------------------|-----------------|----------------|---------------|------------------|
| 65305 | 25265582 | The Goddess of Buttercups and Daisies  |                 | 6.3541         | 3.61          | 24               |
| 13894 | 33605578 | The Forgetting (The Forgetting #1)     |                 | 6.0914         | 3.98          | 8                |
| 7252  | 872302   | E Pluribus Unum                        |                 | 6.037          | 4.02          | 599              |
| 46201 | 15985353 | Heart of Obsidian (Psy-Changeling #12) |                 | 6.0299         | 4.51          | 163              |
| 25743 | 1058174  | The Tough Guide to Fantasyland         |                 | 6.018          | 3.94          | 108              |



# Model Evaluation



## ReadMate: Personalized Book Recommendation System

Big Data Analytics and Recommendation System using Apache Spark

### Model Evaluation

Model evaluation measures how well the ALS model predicts user ratings. RMSE is used as the main evaluation metric.

```
Full reviews count: 3311062
Training sample count: 331765
Sample fraction: 0.1
ALS RMSE: 1.407414863313601
Model: Spark Mllib ALS Collaborative Filtering
```

### RMSE Explanation

RMSE measures the difference between actual user ratings and predicted ratings. A lower RMSE value indicates better prediction accuracy.

### Evaluation Summary

- The ALS model is evaluated using test rating data.
- Predicted ratings are compared with actual ratings.
- RMSE is calculated to measure prediction error.
- The trained model is then used to generate Top-N recommendations.

### Evaluation Metric

- RMSE

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (Predicted_i - Actual_i)^2}{N}}$$

# Thank You



ReadMe